NNN-B-788a
October 25, 1962

SUPERSEDING
Int. Fed. Spec. NNN-B-00788 (ARMY-CmlC)
June 27, 1961
(See section 6)

FEDERAL SPECIFICATION

BULB, DROPPING, PIPET

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers one type of rubber bulbs to be used to fill pipets.

1.2 **Classification.**

1.2.1. *Capacities.* Dropping bulbs covered by this specification shall be of the following capacities, as specified (see 6.1):

Capacities:

2 Milliliters (ml.)
10 ml.
15 ml.
30 ml.
60 ml.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.** The following documents, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

UU-T-111—Tape; Paper, Gummed (Sealing and Securing).
PPP-B-566—Boxes, Folding, Fiberboard.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-B-676—Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification Rules.

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 **Material.** Dropping bulbs shall be made of new natural rubber, non-blooming throughout.

3.2 **Design.** The dropping bulbs shall be conveniently ovoid shaped compression bulb, with one end closed and the other ending in a neck. Neck shall have a circular opening into interior of bulb, 5 ± 1 millimeters (mm.) in diameter. At midpoint of, and at right angles to long axis there shall be a reinforcing bead integral with body wall. The 2-milliliter (ml.) bulb shall have a reinforcing bead at the neck opening.

3.3 **Dimensions.** The dropping bulbs shall conform to the requirements of table I, as specified.

TABLE I.—*Dimensions*

Capacity, ml.	Length,* mm.	Outside diameter,* mm.
2	40	8**
10	45	27
15	60	32
30	68	37
60	82	46

* Tolerance ± 5 mm. ** Tolerance ± 1 mm.

3.4 **Workmanship.** The dropping bulbs shall be smooth, free from ingrained particles, blisters, or other imperfections or contamination such as grease or oil which might detract from their appearance or serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 have been satisfied.

4.3 **Lotting.** A lot of dropping bulbs shall consist of those bulbs of the same capacity made by one manufacturer and submitted for delivery at one time.

4.4 **Sampling.**

4.4.1 *Nondestructive examination.* Sam-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.

pling shall be conducted in accordance with MIL-STD-105.

4.5 **Examination.** Sample dropping bulbs and level A and level B preparation for delivery requirements shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.5.1 *Classification of defects.*

4.5.1.1 *Dropping bulb.*

Categories	Defects
Critical:	None defined
Major:	*AQL 4.0 percent defective*
101	Capacity incorrect
102	Dimensions incorrect
103	Defects (ingrained particles, bloom, cuts, scratches)
104	Contamination (grease, oil)

4.5.1.2 *Level A and B preparation for delivery (section 5).*

Categories	Defects
Critical:	None defined
Major:	*AQL 2.5 percent defective*
101	Unit and shipping containers not as specified
102	Number of bulbs packaged not as specified
103	Unit or shipping containers damaged
104	Unit or shipping container closure not as specified
105	Marking of unit or shipping containers illegible, incomplete, or incorrect

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packaging.** Packaging shall be level A or C, as specified (see 6.1).

5.1.1 *Level A.* The number of bulbs specified, of like capacity, shall be packaged in a box of appropriate size conforming to PPP-B-566, style I or II, type and class optional; or PPP-B-676, type, class, and style optional. The bulbs shall be in one tier and the boxes provided with full height slot type partitions of the same material as the box. Box closures shall be secured with tape conforming to UU-T-111.

5.1.2 *Level C.* Bulbs shall be packaged in accordance with the manufacturer's commercial practice suitable for shipment from the supply source to the first receiving activity for immediate use.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 *Level A.* Bulbs shall be packed in a shipping container conforming to PPP-B-621, class 2; PPP-B-601, overseas type; or PPP-B-636, class 2, W5c or W5s, style RSC. As far as practical, exterior containers shall be of uniform size and shape, be of minimum cube and tare, and contain identical quantities of like items. Gross weight shall not exceed the weight limitations of the applicable box specification or appendix thereto. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.2.2 *Level B.* Bulbs shall be packed in accordance with 5.2.1 except that shipping containers shall conform to PPP-B-621, class 1; PPP-B-601, domestic type; or PPP-B-636, type I, class 1, style RSC, fabricated from not less than 25-pound test fiberboard.

5.2.3 *Level C.* Bulbs shall be packed to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 **Marking.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Level of packaging and packing required.
(*c*) Capacity of bulb required.
(*d*) Number of bulbs per package required.

6.2 **Military use.** Items procured under this specification for military use are to be limited to the variety shown on the applicable military standards and military supply standards. Personnel of the military departments are requested to refer to these documents for guidance.

6.3 **Supersession data.** This specification covers the requirements of Military Specification MIL-B-2306, dated 24 July 1950.

6.4 **Transportation description.** Transportation description applicable to this item is:

Rubber articles NOIBN.
Carload minimum weight 20,000 pounds, subject to Rule 34, Uniform Freight Classification.

Rubber articles NOI (specify density).
Truckload minimum weights 7,100, 8,600, 10,000, 11,400, 13,700, 15,000, 16,000, 18,000, 21,000, 21,400, 25,700, and 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:
Army—MU
Navy—Md
Air Force—DAY

Other Interests:
Army—E M O

Copies of this specification may be purchased for 5 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.